Day 5 — Charts & EDA

Correlation Task

Complete Step-by-Step Visual Guide

What you will learn in this guide

1. What is correlation — in plain English with real-world examples
2. What the r value (correlation number) means
3. How to build a Correlation Matrix in Excel — step by step with screenshots
4. How to apply a colour heatmap so patterns are instantly visible
5. How to build a Feature R^2 Ranking chart to find your best predictors
6. How to read and interpret everything you built
7. Common mistakes and how to avoid them

■ 30-Second Summary

Correlation measures **how strongly two columns move together**. When Study Hours goes up, does Score go up too? Correlation tells you how tight that relationship is — and gives you a number between -1 and +1. A Correlation Matrix shows this number for EVERY pair of columns at once. With a colour heatmap applied, you can spot your strongest predictors in 2 seconds.

SECTION 1 — What Is Correlation?

Correlation is a number that tells you **how strongly two things are related**. It answers the question: "When X increases, does Y tend to increase, decrease, or neither?"

Positive Correlation (+)	Negative Correlation (–)	No Correlation (≈ 0)
When one goes UP, the other goes UP too.	When one goes UP, the other goes DOWN.	No pattern — changes are random.
■■ Hot weather → more ice cream sold ■ More study → higher score ■ More exercise → lower weight	❄■ Cold weather → more heating bills ■ More absences → lower grade ■ Faster driving → less travel time	■ Shoe size → exam score ■ Birthday month → salary ■ TV brand → height
r close to +1	**r close to -1**	**r close to 0**

The r Value — What Each Number Means

r Value	Strength	Direction	Real Example
+0.9 to +1.0	Very Strong	Both rise together	Height & weight
+0.5 to +0.9	Moderate–Strong	Both rise	Study hours & exam score
+0.1 to +0.5	Weak	Barely related	Sleep hours & score
0 to ±0.1	None / Negligible	No pattern	Shoe size & salary
-0.1 to -0.5	Weak Negative	Slight opposite	Price & demand
-0.5 to -0.9	Moderate–Strong Neg	Opposite	Absences & grade
-0.9 to -1.0	Very Strong Neg	Perfect opposite	Speed & travel time

■ Key Formula

In Excel: =CORREL(column1, column2) Example: =CORREL(B2:B81, C2:C81) This gives you r — a single number between -1 and +1. To get R-squared (how much % of variation is explained): =RSQ(column1, column2) Or simply: =CORREL(col1,col2)^2

SECTION 2 — What Is a Correlation Matrix?

A Correlation Matrix shows the correlation between **every pair of columns** in your dataset at the same time. Instead of running =CORREL() 16 separate times for 4 features, the matrix gives you all 16 numbers in one organised table.

What a 4×4 Correlation Matrix looks like:

	Score	StudyHours	Attendance	Income
Score	**1.00**	**+0.82**	**+0.61**	**+0.04**
StudyHours	**+0.82**	**1.00**	**+0.58**	**+0.02**
Attendance	**+0.61**	**+0.58**	**1.00**	**-0.03**

Income	**+0.04**	**+0.02**	**-0.03**	**1.00**

Reading the matrix: Each cell = CORREL(row feature, column feature). The diagonal is always 1.0 (a feature perfectly correlates with itself). The matrix is symmetric — the top-right mirrors the bottom-left.

How to read the insights from this matrix:

✓	StudyHours vs Score: r=+0.82	STRONG positive predictor — include in your model
✓	Attendance vs Score: r=+0.61	MODERATE predictor — useful, include it
■	StudyHours vs Attendance: r=+0.58	Moderate correlation between features — watch for multicollinearity
✗	Income vs Score: r=+0.04	NO relationship — drop Income from the model

SECTION 3 — Build the Correlation Matrix in Excel (Step by Step)

■ Before You Start

Open your Day5_EDA.xlsx workbook (or any workbook with a Table of data). You need at least 4 numeric columns — for example: Score, StudyHours, Attendance, Income. Create a new sheet called 'CorrelLab' for this exercise. All steps below happen on this new sheet.

1

Set Up the Matrix Headers

In your CorrelLab sheet:
• Type your feature names as ROW headers in column A: A2=Score, A3=StudyHours, A4=Attendance, A5=Income
• Type the same feature names as COLUMN headers in row 1: B1=Score, C1=StudyHours, D1=Attendance, E1=Income
• Bold all headers. Make them the same as your Table column names exactly.

2

Enter the CORREL Formula in the First Cell

Click cell B2 (row = Score, column = Score — the diagonal).
Type: =CORREL(EDAData[Score], EDAData[Score])
Press Enter. You get 1.00 — a feature always perfectly correlates with itself.

Now click cell B3 (row = StudyHours, column = Score).
Type: =CORREL(EDAData[StudyHours], EDAData[Score])
This gives you r between StudyHours and Score.

3 Fill the Entire 4x4 Matrix

For each cell, the formula is =CORREL(row_feature_column, col_feature_column).
The pattern is always the same — the row name tells you the first array, the column name tells you the second.

Fill all 16 cells (4 rows × 4 columns). The diagonal will always be 1.00.
TIP: The matrix is symmetric — B3 should equal C2 (same pair, different order).
Format all 16 cells to 2 decimal places: Ctrl+1 → Number → 2 decimal places.

4 Apply the Colour Heatmap (Conditional Formatting)

Select the 16 value cells (B2:E5) — do NOT include headers.

Home tab → Conditional Formatting → Color Scales → More Rules.
Set a 3-Color Scale:
• Minimum value: -1 → Red (#FF4444)
• Midpoint value: 0 → White (#FFFFFF)
• Maximum value: +1 → Green (#00AA66)

Click OK. The matrix is now a visual heatmap!

5 Dim the Weak Correlations (Optional But Powerful)

Add a second rule to grey-out correlations that are too weak to matter.

Still with B2:E5 selected → Conditional Formatting → New Rule →
'Format only cells that CONTAIN' → Cell Value → Between → -0.1 and 0.1
Set font colour to light grey (#AAAAAA).

Now weak correlations fade into the background, and strong ones stand out clearly.

6 Run ToolPak Correlation (Automatic Method)

For a faster alternative:
Data tab → Data Analysis → Correlation
Input Range: select all 4 feature columns WITH headers
Check: 'Labels in first row'
Output Range: pick an empty cell
Click OK.

Excel fills the matrix automatically! Apply the same colour scale to this output.
Compare both methods — the numbers should be identical.

■ What Your Finished Matrix Should Show

After applying the heatmap: dark green cells = strong positive relationship. Dark red cells = strong negative relationship. White or grey cells = no relationship. The diagonal (top-left to bottom-right) = always bright green (1.00). If a row or column is mostly white/grey, that feature has no relationship with others — consider dropping it.

SECTION 4 — Feature R² Ranking Chart (Most Important Chart)

The R² Ranking Chart answers the most important ML question: **which features are the best predictors of my target?** It takes the correlation between each feature and the target, squares it, and plots them as a bar chart from strongest to weakest. This is your feature selection tool.

Understanding R² (R-squared):

r (correlation)	R² = r × r	Plain English Meaning
+0.82 (StudyHours)	`0.82 × 0.82 = 0.67`	Study hours explains 67% of score variation
+0.61 (Attendance)	`0.61 × 0.61 = 0.37`	Attendance explains 37% of score variation
+0.04 (Income)	`0.04 × 0.04 = 0.002`	Income explains only 0.2% — essentially nothing

How to Build the R² Ranking Table and Chart:

1

Build the R² Table

In your CorrelLab sheet, create a 3-column table:
Column A: Feature name (StudyHours, Attendance, Income)
Column B: r value — =CORREL(EDAData[Feature], EDAData[Score])
Column C: R² value — =B2^2 (or =RSQ(EDAData[Score], EDAData[Feature]))

Fill for all 3 features. Column C is your feature importance score.

2

Sort by R² Descending

Click anywhere in column C → Data → Sort Largest to Smallest.
Your most important predictor is now at the top.

This sorted order is exactly how scikit-learn displays feature importance in Python.
The feature at the top = the one your model should definitely include.

3

Add Data Bars to the R² Column

Select your R² values (column C) → Home → Conditional Formatting → Data Bars → Solid Fill → Green.
In the 'More Rules' option, set Maximum = 1.0 (so bars are proportional to a perfect R²=1).

Now each R² cell shows a mini bar inside it — a sparkline of feature importance.
This is cleaner and more informative than a separate chart.

4 Build the Feature Importance Bar Chart

Select Feature Name column and R^2 column together (Ctrl+click to select both).
Insert → Bar Chart → 2D Bar (horizontal bars — easier to read with text labels).
Chart Design → Switch Row/Column if needed so features are on the Y-axis.
Title: 'Feature Predictive Power (R^2 with Target Score)'

5 Format the Chart Professionally

- Sort bars from highest to lowest (select data → sort before charting)
- Delete the chart legend (not needed — only one data series)
- Delete or lighten gridlines
- Add data labels: right-click bars → Add Data Labels → format to 2 decimal places
- Colour bars: dark green for strong predictors, light green for weak
- Add a vertical reference line at R^2=0.1 (your drop threshold)

Example — What Your R^2 Chart Shows:

Feature	r	R^2	Bar	Decision
StudyHours	+0.82	0.67	■■■■■■■■■■■■■■■■■■■■ 67%	✓ Include
Attendance	+0.61	0.37	■■■■■■■■■■ 37%	✓ Include
Income	+0.04	0.002	0.2%	✗ Drop

SECTION 5 — Multicollinearity Check (Critical for ML)

After finding which features predict your TARGET well, you must also check if any two **features predict each other** too well. This is called **multicollinearity** — and it damages regression models.

■■ What Is Multicollinearity?

If StudyHours and Attendance have r=0.85, they're almost measuring the same thing (students who study more also attend more). Including BOTH in your model confuses it — it can't tell which feature is doing the work. The model becomes unstable and coefficient values become unreliable. Rule: If two FEATURES (not target) have |r| > 0.7 with each other — that is a multicollinearity warning. Consider dropping the weaker one (the one with lower R^2 with the target).

How to Check Multicollinearity from Your Matrix:

1. Look at your correlation matrix — but IGNORE the row and column for your TARGET (Score).
2. Focus only on the FEATURE-to-FEATURE cells (StudyHours vs Attendance, StudyHours vs Income, etc.).
3. Flag any cell where the absolute value |r| > 0.7 — these are your multicollinearity pairs.
4. For each flagged pair: keep the feature with HIGHER R^2 with the target. Drop the other.
5. After dropping, re-run the matrix to confirm no remaining |r| > 0.7 between features.

Matrix Cell	r Value	Type	What To Do
Score vs StudyHours	0.82	Target-Feature	Good predictor — keep StudyHours
Score vs Income	0.04	Target-Feature	No relationship — drop Income
StudyHours vs Attendance	0.85	Feature-Feature	**Multicollinearity! Drop weaker one**
StudyHours vs Income	0.06	Feature-Feature	Fine — no multicollinearity problem

SECTION 6 — Common Mistakes & How to Avoid Them

■ Mistake 1: Including headers in the CORREL range

Wrong: =CORREL(A1:A81, B1:B81) if A1 and B1 are text headers
Right: =CORREL(A2:A81, B2:B81) — start from row 2 (first data row)
Better: =CORREL(EDAData[Score], EDAData[StudyHours]) — use Table structured references

■ Mistake 2: Applying colour scale to the header cells

Only select the 16 VALUE cells (e.g. B2:E5) — NOT the headers in row 1 or column A.
If you include headers, the colour scale gets confused by text values.

■ Mistake 3: Thinking r=0.5 is 'weak'

r=0.5 means R^2=0.25 — your feature explains 25% of the target's variation.
In real-world messy data, r=0.5 is often a VERY useful predictor.
Only drop features below |r|=0.1 (R^2<0.01 = less than 1% explained).

■ Mistake 4: Confusing feature-target r with feature-feature r

High r between a FEATURE and the TARGET = good (useful predictor).
High r between two FEATURES = bad (multicollinearity problem).
They look the same in the matrix but mean opposite things.

■ Mistake 5: Forgetting that r only measures LINEAR relationships

r=0.02 doesn't mean the feature is useless — it means there's no LINEAR relationship.
There could still be a curved (non-linear) relationship that correlation misses.
Always look at a scatter plot alongside the correlation number.

SECTION 7 — Quick Reference Cheat Sheet

Task	Excel Formula / Action

Correlation between 2 columns	`=CORREL(EDAData[Col1], EDAData[Col2])`
R-squared (predictive power)	`=RSQ(EDAData[Target], EDAData[Feature])`
Auto-fill matrix (ToolPak)	`Data → Data Analysis → Correlation`
Apply heatmap colours	`Select values → Conditional Format → Color Scales → More Rules → 3`
Set colour scale values	`Min=-1 Red, Mid=0 White, Max=+1 Green`
Dim weak correlations	`Conditional Format → New Rule → Between -0.1 and 0.1 → Grey font`
Add data bars to R² column	`Select R² values → Conditional Format → Data Bars → Solid Fill → G`
Build feature importance chart	`Select Feature + R² columns → Insert → Bar Chart → 2D Horizontal Ba`
Sort chart by importance	`Sort R² table descending before charting`
Check multicollinearity	`Feature-Feature |r| > 0.7 = problem → drop weaker feature`

■ You Are Now Ready

After completing this task in Excel you will have: a colour-coded correlation heatmap, a feature R^2 ranking bar chart, multicollinearity flags, and a clear list of which features to include in your Day 6 regression model. That list of features is one of the most important outputs of the entire 10-day programme — and you just learned to build it from scratch, step by step.